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UNIVERSITY

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PROGRAMMING WITH PYTHON

Lekha A

Computer Applications

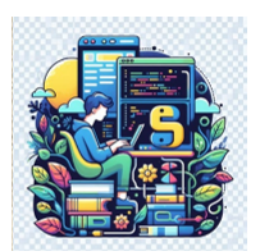
PROGRAMMING WITH PYTHON

Object-Oriented Programming (OOP) and Advanced Concepts

Introduction to OOP

Lekha A

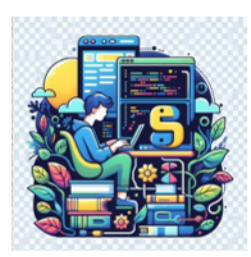
Computer Applications



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Introduction

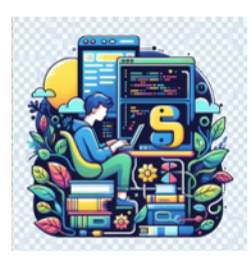
- Everyone knows what an object is?
 - A tangible thing that can be sensed, felt and manipulated.
- Software objects may not be tangible things that can be picked up, sensed or felt, but they are models of something that can do certain things and have certain things done to them.



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Introduction

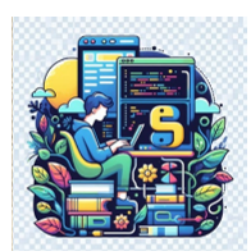
- Formally, an object is a collection of data and associated behaviors.
- What does it mean to be object-oriented?
 - In the dictionary, oriented means directed toward.
- Object-oriented means functionally directed toward modeling objects.



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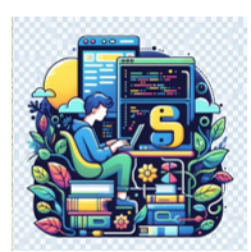
- Object-oriented programming is one of the most effective approaches to writing software.
- In object-oriented programming classes are written that represent real-world things and situations, and objects are created based on these classes.



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Introduction

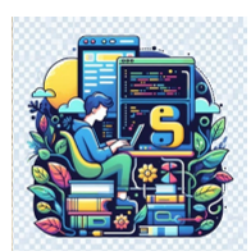
- When a class is written, the general behavior that a whole category of objects can have is defined.
- When individual objects are created from the class, each object is automatically equipped with the general behavior; and each object will have whatever unique traits the user desires.



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Introduction

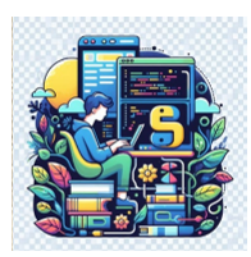
- This is one of many techniques used for modeling complex systems.
- It is defined by describing a collection of interacting objects via their data and behavior.



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What is OOP?

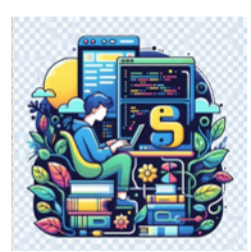
- Object-oriented programming (OOP) is a programming paradigm that uses objects and their interactions to design applications and computer programs.
- Objects are self-contained entities that contain both data and code.
- They can interact with each other by sending and receiving messages.



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Advantages of OOP

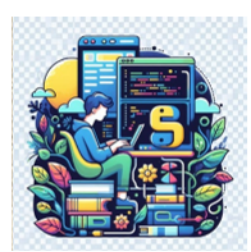
- OOP has several advantages over other programming paradigms, such as procedural programming
 - Modularity
 - Objects can be created and managed independently, which makes code more modular and reusable.



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Advantages of OOP

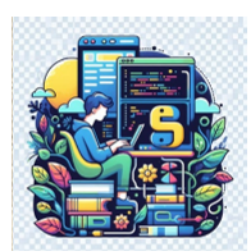
- Encapsulation
 - Objects can encapsulate their data and code, which makes them more secure and easier to maintain.
- Inheritance
 - Objects can inherit properties and behaviors from other objects, which allows for code reuse and reduces duplication.



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Advantages of OOP

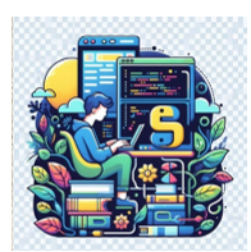
- Polymorphism
 - Objects can respond to different messages in different ways, which makes code more flexible and adaptable.



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Aim of OOP

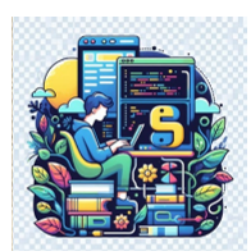
- The main aim of OOP is to bind together the data and the functions that operate on them so that no other part of the code can access this data except that function.
- This paradigm closely resembles how humans perceive objects in real life, thus reducing the complexity associated with software development.



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Definitions of terms

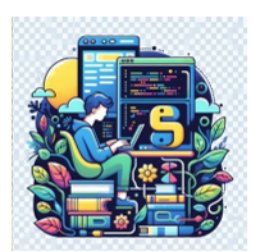
- **Class**
 - A user-defined prototype for an object that defines a set of attributes that characterize any object of the class.
 - The attributes are data members (class variables and instance variables) and methods, accessed via dot notation.



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Definitions of terms

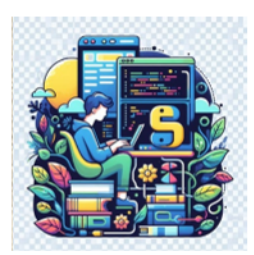
- **Class**
 - Classes are blueprints or templates for creating objects.
 - They define the structure (attributes) and behavior (methods) that objects of that class will have.



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Definitions of terms

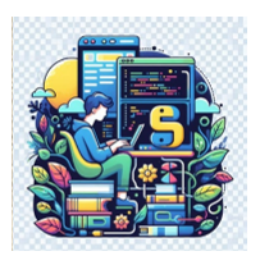
- **Object**
 - A unique instance of a data structure that's defined by its class.
 - An object comprises both data members (class variables and instance variables) and methods.



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Definitions of terms

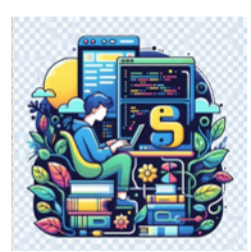
- **Object**
 - Objects created from the same class share common characteristics.
 - They encapsulate both data (attributes or properties) and the operations (methods or functions) that can be performed on that data.



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Definitions of terms

- **Object**
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Definitions of terms

- **Class**

```
|: class Vehicle:
    def __init__(self, make, model, year):
        self.make = make
        self.model = model
        self.year = year

    def move(self):
        pass
```

```
|: class Car(Vehicle):
    def drive(self):
        print(f"The {self.make} {self.model} {self.year} is driving.")
```

```
|: class Truck(Vehicle):
    def haul(self):
        print(f"The {self.make} {self.model} {self.year} is hauling cargo.")
```

```
my_car = Car("Toyota", "Camry", 2023)
```

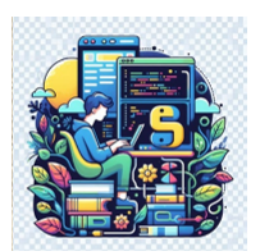
```
my_truck = Truck("Ford", "F-150", 2023)
```

```
my_car.drive()
```

The Toyota Camry 2023 is driving.

```
my_truck.haul()
```

The Ford F-150 2023 is hauling cargo.

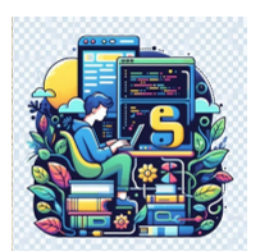


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Comparing OOP with Procedural Programming

• OOP

Focus	Objects and their interactions
Data and code	Data and code encapsulated within objects
Modularity	Objects are self-contained and reusable
Code reuse	Objects and procedures can be reused
Maintenance	Code is easier to maintain because changes to one object do not affect other objects
Complexity	OOP is well-suited for complex applications

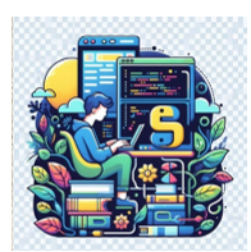


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Comparing OOP with Procedural Programming

Procedural Programming

Focus	Steps or procedures
Data and code	Separate
Modularity	Procedures are self-contained but not always reusable
Code reuse	Procedures can be reused, but code reuse is not as central to procedural programming as it is to OOP



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Comparing OOP with Procedural Programming

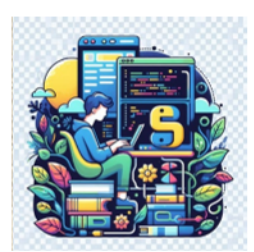
Procedural Programming

Maintenance

Code can be more difficult to maintain because changes to one procedure can affect other procedures

Complexity

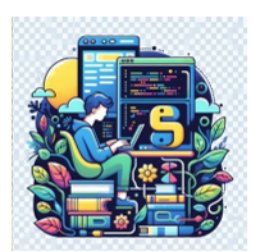
Procedural programming is well-suited for simpler applications



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Uses of OOP's

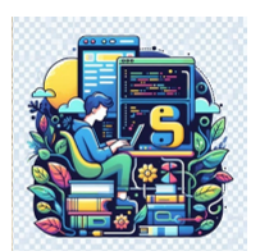
- OOP is used in a wide variety of software applications, including
 - Operating systems (e.g., Windows, macOS, Linux)
 - Web applications (e.g., Google Search, Facebook, Amazon)
 - Mobile applications (e.g., iOS apps, Android apps)
 - Desktop applications (e.g., Microsoft Office, Adobe Photoshop, Autodesk Maya)



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Uses of OOP's

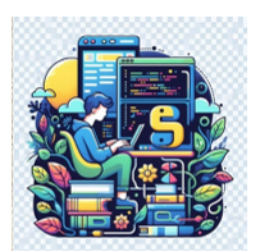
- OOP is used in a wide variety of software applications, including
 - Embedded systems (e.g., car navigation systems, medical devices, industrial control systems)
 - Video games (e.g., Fortnite, Grand Theft Auto V, The Witcher 3: Wild Hunt)



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Uses of OOP's

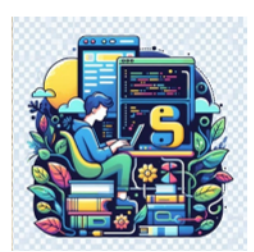
- In an operating system
 - OOP is used to model the different components of the system, such as files, processes, and threads.
 - This allows the system to be more modular and extensible, as new components can be added without having to modify existing code.



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Uses of OOP's

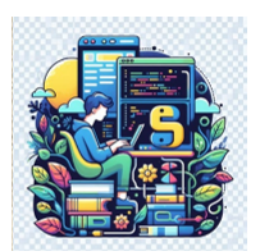
- In a web application
 - OOP is used to model the different pages and components of the application, such as user profiles, product listings, and shopping carts.
 - This allows the application to be more scalable and maintainable, as new features can be added without having to rewrite the entire application.



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Uses of OOP's

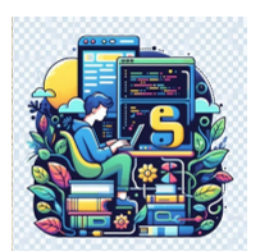
- In a mobile application
 - OOP is used to model the different views and controllers of the application, such as the login screen, the main product list, and the checkout screen.
 - This allows the application to be more responsive and user-friendly, as new views and controllers can be added without having to modify existing code.



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Uses of OOP's

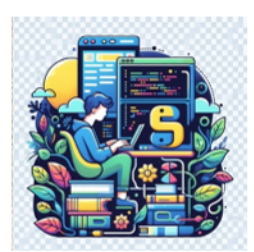
- In an embedded system
 - OOP is used to model the different devices and sensors of the system, such as a temperature sensor, a GPS receiver, or a motor controller.
 - This allows the system to be more adaptable to different hardware configurations, as new devices and sensors can be added without having to modify existing code.



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Uses of OOP's

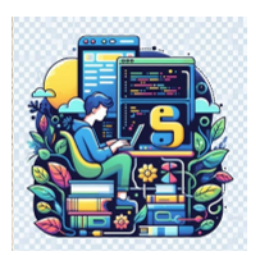
- In a desktop application
 - OOP is used to model the different documents and workspaces of the application, such as a Word document, an Excel spreadsheet, or a Photoshop image.
 - This allows the application to be more powerful and flexible, as new documents and workspaces can be added without having to modify existing code.



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Uses of OOP's

- In a video game
 - OOP is used to model the different objects in the game world, such as the player character, the enemies, and the environment.
 - This allows the game to be more realistic and immersive, as new objects and features can be added without having to rewrite the entire game.



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Recap

- Introduction
- What is OOP?
- Advantages
- Aim of OOP
- Class
- Object
- Comparison with Procedural Programming
- Uses of OOP



THANK YOU

Lekha A
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